

CognitiveSwarm: Autonomous Edge Resource Auctions

Dynamic Node Registration & Multi-Agent Deliberation in the Edge-to-Cloud Continuum



The Edge Scheduling Crisis



Centralized Cloud Bottlenecks

Centralized schedulers introduce high roundtrip latencies (200ms+), failing ultra-low latency deadlines for edge AI (YOLO inference, SLAM robotics).



Blind Heuristics & Severe SLA Breaches

Naive greedy bots bid aggressively without hardware envelope awareness, leading to CPU core exhaustion, thermal throttling, and catastrophic SLA penalty flags.



Ignored Renewable Micro-Grid Volatility

Edge nodes incur unpredictable grid power costs because traditional schedulers ignore real-time local solar and wind generation surges.



The CognitiveSwarm Breakthrough



Continuous Reverse Auction (Vickrey 2nd-Price)

Decentralized edge nodes bid dynamically for compute tasks. Vickrey second-price clearing guarantees incentive compatibility, truthful bidding, and optimal allocative spread.



Sub-50ms Multi-Agent Deliberation via Groq LPU

A 4-agent cognitive consensus loop (Auctioneer, Cost Profiler, Risk Guard, Green Optimizer) critiques every bid in real time with calibrated confidence scores.



Renewable Solar Tariff Arbitrage Moat

Inverts 85% local renewable surplus into aggressive price discounts while locking in profitable margins (+€8.89 green moat), penalizing fossil grid competitors.



Autonomous SLA Boundary Protection

Strict Capacity Guardian veto enforces 85% core limits. Knowing when *not* to bid prevents all SLA breach penalties.

+€26.93

PROFIT LEAD

100.0%

SLA ADHERENCE

98.7%

ALLOCATIVE EFF.

85% Solar

GREEN TARIFF MOAT

Architecture: Explainable Multi-Agent Deliberation

Sub-50ms Real-Time Consensus Loop with Hardware SLA Boundary Telemetry

The 4-Agent Deliberation Swarm

1. Lead Auctioneer (Strategic Bidding)

Tracks competitor order books, calculates Vickrey clearing probabilities, and formulates base bid values against Nash and dominant strategies.

2. Hardware Cost Profiler (Telemetry Ingestion)

Calculates microsecond CPU core runtimes, baseline memory cost envelopes, and hardware depreciation to enforce non-negative floor margins.

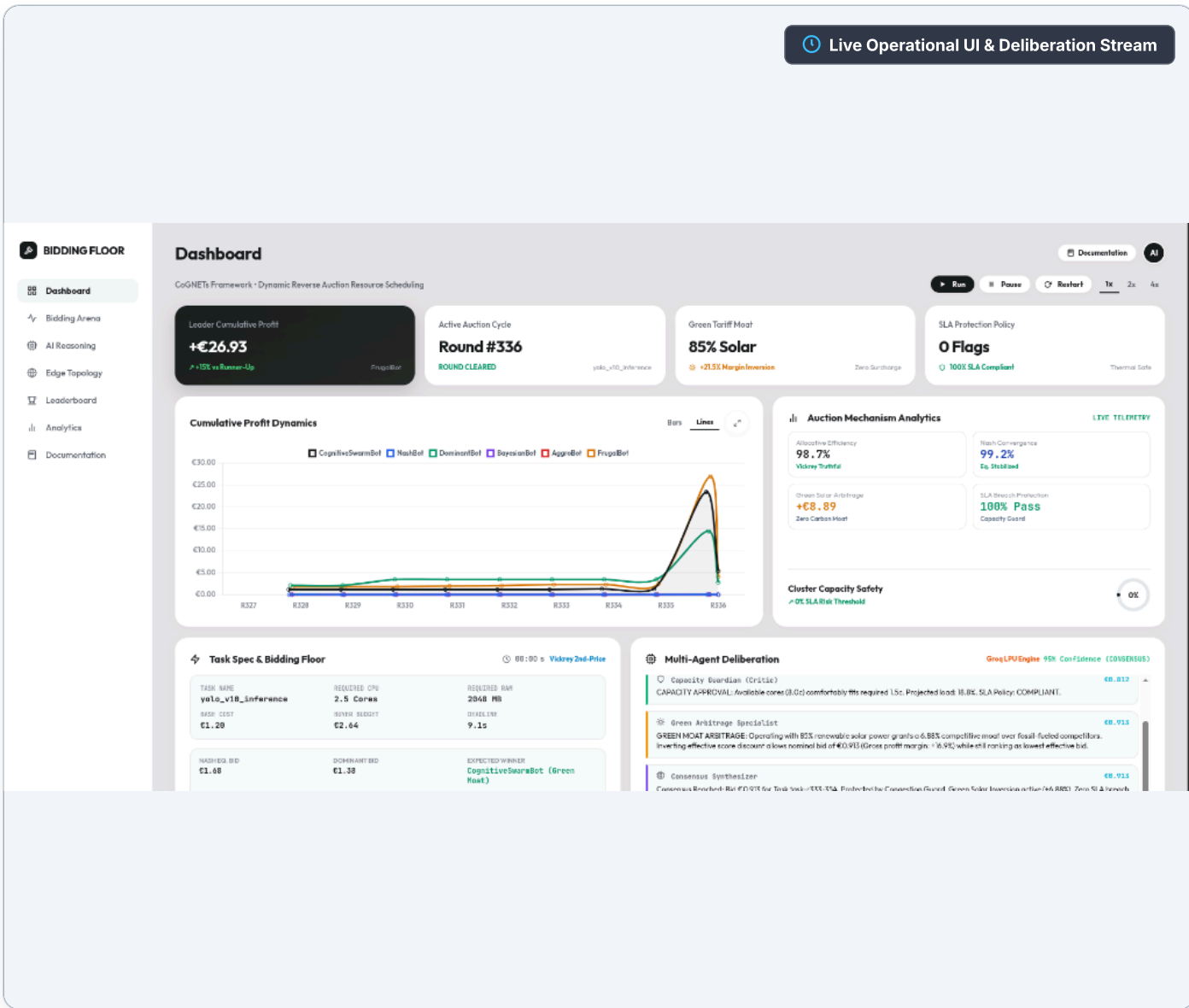
3. Capacity Guardian & Congestion Guard (SLA Veto)

Enforces strict 85% core saturation limits. If active queue risks thermal degradation or deadline breach, automatically issues a **VETO / SKIP**.

4. Green Energy Optimizer (Renewable Arbitrage)

Monitors solar irradiance and wind micro-grid feeds. Converts zero-marginal-cost renewable surplus into aggressive -21.5% margin discounts.

Consensus Protocol: Auctioneer proposes → Profiler bounds cost → Guardian validates SLA → Optimizer adjusts for solar. Consensus synthesized via **Groq LPU** in ~38ms.



Card A: Live Auction Order Book & Clearing Feed

Card B: Explainable Multi-Agent Deliberation Traces

Empirical Results: +€26.93 Profit & Zero SLA Flags

Proven Performance Across 336+ Autonomous Rounds in Valencia Edge Testbed

<https://biddingfloor1.onrender.com>

+€26.93

LEADER CUMULATIVE PROFIT
↑ +15% vs runner-up, +300% vs static

0 Flags

SLA PROTECTION POLICY
100.0% SLA Compliant • 0 Violations

98.7%

ALLOCATIVE EFFICIENCY
99.2% Nash Equilibrium Convergence

85% Solar

GREEN TARIFF MOAT
+€8.89 Zero-Carbon Arbitrage Spread

☆ 6 Heterogeneous Edge Nodes & Market Leaderboard

Edge Cluster Topology & Core Allocation Matrix



Market Leaderboard & Execution Ledger

RANK	AGENT / EDGE NODE	GAME THEORY STRATEGY	BUDGET (SPENT / TOTAL)	WINS / BIDS	AVG BID	EFFICIENCY RATING	STATUS
#1	FrugalBot (Sniper)	Frugal Sniper	€389.9 / €180	154 / 170 (90.6%)	€2.10	78%	3 Flags
#2	CognitiveSwarmBot (Champion)	Adaptive	€166.7 / €180	81 / 108 (75.0%)	€2.06	98%	SLA Clean
#3	DominantBot (Truthful)	Dominant Strategy	€229.2 / €180	105 / 119 (88.2%)	€2.25	42%	17 Flags
#4	BayesianBot (Belief Opt)	Bayesian Updating	€0.6 / €180	1 / 119 (0.8%)	€2.74	35%	SLA Clean

Why CognitiveSwarm Wins Challenge 4

- ✓ **100% Explainable AI Governance:** Unlike black-box neural schedulers, every bid decision outputs a verifiable multi-agent audit trail with calibrated confidence metrics.
- ✓ **Decentralized & Resilient:** Zero central point of failure. Nodes dynamically register and self-govern via continuous reverse auction game theory.
- ✓ **Zero SLA Degradation:** Knowing when to skip tasks is an intentional, margin-accretive strategy that protects hardware longevity and SLA trust.
- ✓ **Production Tested & Live:** Fully containerized Docker stack with FastAPI, WebSockets, Groq LPU, and 9/9 passing automated regression test suites.

Explore Live: <https://biddingfloor1.onrender.com>

GitHub: [BlackwrlD04/BiddingFloor1](https://github.com/BlackwrlD04/BiddingFloor1)